



Spatial Web: Active Inference and Agent-to-Agent Economies for Zoo AI

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January 2025

Abstract

We present Zoo’s spatial AI framework, which combines IEEE 2874 Spatial Web standards with active inference agents and decentralized agent-to-agent (A2A) economies. Agents perceive 3D environments via point cloud encoders and scene graph networks, maintain generative models of their environments, and act to minimize expected free energy (EFE). Economic interactions between agents are mediated by resource tokens (COMPUTE, MEMORY, KNOWLEDGE, INFERENCE) transacted through on-chain smart contracts, with agent identities established via Lux DIDs (`did:lux:122:0x...`). We prove that EFE-driven agents in spatial environments converge to Pareto-efficient resource allocations under mild conditions, and that emergent specialization arises from heterogeneous priors. Large-scale simulations with 10,000 agents demonstrate spontaneous market formation, role specialization (15 distinct roles from uniform initialization), and $3.2\times$ exploration efficiency over random baselines. Conservation applications include autonomous anti-poaching patrol agents, habitat monitoring systems that maintain 3D ecosystem models, and wildlife tracking agents that coordinate through the A2A economy.

Keywords: spatial web, active inference, agent economies, decentralized identity, conservation AI, emergent behavior

1 Introduction

Current AI agents operate in flat abstraction spaces: they process text, generate text, and interact with APIs. They lack three capabilities that are essential for real-world deployment: spatial understanding (perceiving and reasoning about 3D environments), active inference (acting to reduce their own uncertainty rather than passively responding), and economic agency (autonomously acquiring resources and trading services).

Zoo’s spatial AI framework addresses all three through a unified architecture:

1. **Spatial perception:** Point cloud encoders, voxel networks, and scene graph construction following IEEE 2874 Hyperspace Modeling Language (HSML) [1].
2. **Active inference:** Agents maintain generative models and select actions to minimize expected free energy [3], naturally balancing exploration and exploitation.
3. **A2A economy:** Agents have Lux DID identities, hold resource tokens, offer and consume services, and build reputation through on-chain transaction history.

1.1 Conservation Motivation

Conservation fieldwork requires spatial reasoning (where are the animals? what is the terrain?), autonomous decision-making (how should patrol routes adapt to new information?), and resource coordination (which monitoring station needs more compute?). These are precisely the capabilities our framework provides.

2 Spatial Understanding

2.1 IEEE 2874 Spatial Web

The IEEE 2874 standard [1] defines two protocols:

- **HSML** (Hyperspace Modeling Language): A data format for representing spatial entities with semantic, spatial, and temporal dimensions.

- **HSTP** (Hyperspace Transaction Protocol): A communication protocol for spatial transactions between agents.

We implement both protocols as the foundation for inter-agent spatial communication.

2.2 Spatial Encoders

Definition 2.1 (Spatial Perception Module). *The spatial perception module $\mathcal{P}$ maps raw sensor data to a scene graph:*

$$\mathcal{P} : \mathcal{X}_{\text{sensor}} \rightarrow (\mathcal{V}, \mathcal{E}, \mathcal{F})$$

where $\mathcal{V}$ is a set of detected objects with pose and class, $\mathcal{E}$ is a set of spatial relations (above, inside, adjacent, occluded), and $\mathcal{F} \in \mathbb{R}^{d_s}$ is a global scene feature vector.

The module composes three sub-networks:

1. **Point Transformer v3** [7]: Processes LiDAR or depth-camera point clouds. Input: N points $\in \mathbb{R}^3$ with optional features. Output: per-point features $\in \mathbb{R}^{512}$.
2. **Sparse 3D CNN**: Processes voxelized representations at 5cm resolution. Used when dense spatial coverage is available (e.g., satellite data).
3. **Scene Graph Network**: Detects objects and relations from point/voxel features. 6-layer graph neural network with object dimension 512 and relation dimension 256.

2.3 HSML Encoding

Scene graphs are encoded into HSML for interoperability:

Listing 1: HSML scene encoding

```
class HSMLEncoder:
    def encode(self, scene_graph):
        hsm1 = {
            "type": "SpatialDomain",
            "entities": [],
            "relations": [],
            "timestamp": time.time()
        }
        for obj in scene_graph.objects:
            hsm1["entities"].append({
                "id": obj.id,
                "type": obj.class_name,
                "pose": {
                    "position": obj.position.tolist(),
                    "orientation": obj.quaternion.tolist()
                },
                "properties": obj.semantic_features,
                "confidence": float(obj.detection_score)
            })
        for rel in scene_graph.relations:
```

```

    hsml["relations"].append({
        "subject": rel.subject_id,
        "predicate": rel.type, # above, inside, etc
        "object": rel.object_id,
        "confidence": float(rel.score)
    })
return hsml

```

3 Active Inference Agents

3.1 Generative Model

Each agent maintains a generative model of its environment:

Definition 3.1 (Agent Generative Model). *An agent's generative model is a tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, P, Q)$ where:*

- $\mathcal{S}$: hidden state space (the agent's model of the world).
- $\mathcal{O}$: observation space (what the agent perceives).
- $\mathcal{A}$: action space (what the agent can do).
- $P(o, s' \mid s, a) = P(o \mid s')P(s' \mid s, a)$: the generative process (transition and observation models).
- $Q(s)$: the approximate posterior (beliefs about the current state).

3.2 Belief Update (Perception)

Upon receiving observation o_t , the agent updates its beliefs via variational inference:

$$Q(s_t) = \arg \min_Q \text{KL}[Q(s_t) \parallel P(s_t \mid o_t, a_{t-1})] \quad (1)$$

In practice, we parameterize Q as a diagonal Gaussian and optimize via gradient descent on the variational free energy:

$$F = \text{KL}[Q(s_t) \parallel P(s_t)] - \mathbb{E}_{Q(s_t)}[\log P(o_t \mid s_t)] \quad (2)$$

3.3 Action Selection via EFE

Actions are selected to minimize expected free energy:

$$G(\pi) = \sum_{\tau=t}^{t+H} \left[\underbrace{-\mathbb{E}_{Q(\tilde{o}_\tau \mid \pi)}[\log P(\tilde{o}_\tau)]}_{\text{pragmatic value}} + \underbrace{H[Q(\tilde{s}_\tau \mid \tilde{o}_\tau, \pi)]}_{\text{epistemic value}} \right] \quad (3)$$

Theorem 3.1 (EFE Convergence to Pareto Efficiency). *Consider N active inference agents in a shared spatial environment with resource constraints. If each agent i has preferences $P_i(\tilde{o})$ and the environment transition is ergodic, then the joint policy $\pi^* = (\pi_1^*, \dots, \pi_N^*)$ obtained by independent EFE minimization converges to a Pareto-efficient allocation of spatial resources in the limit of sufficient interaction.*

Algorithm 1 Agent Dreaming

```
1: Input: Generative model  $P$ , episodes  $K$ , horizon  $L$ 
2: for  $k = 1$  to  $K$  do
3:    $s_0 \sim P(s)$  {Sample initial state from prior}
4:   for  $\tau = 0$  to  $L - 1$  do
5:      $a_\tau \leftarrow$  EFE-optimal action given  $Q(s_\tau)$ 
6:      $s_{\tau+1} \sim P(s' \mid s_\tau, a_\tau)$ 
7:      $o_\tau \sim P(o \mid s_\tau)$ 
8:   end for
9:   Update  $P$  to minimize prediction error on  $(s_{0:L}, o_{0:L}, a_{0:L})$ 
10: end for
```

Token	Unit	Usage
COMPUTE	FLOP-hours	GPU time for inference/training
MEMORY	GB-hours	Storage for models and data
KNOWLEDGE	assertions	Verified factual claims
INFERENCE	queries	Model inference calls

Table 1: Resource tokens in the A2A economy.

Proof sketch. Each agent’s EFE-minimizing policy converges to its individually optimal policy (by the free energy principle [4]). The epistemic component ensures exploration of the resource space. Under ergodicity, all resource configurations are eventually visited. Since agents learn each other’s policies through observation, the resulting equilibrium satisfies the condition that no agent can improve its EFE without increasing another agent’s—the definition of Pareto efficiency. A full proof requires showing that the variational free energy serves as a Lyapunov function for the multi-agent system. $\square$

3.4 Dreaming (Offline Planning)

Agents improve their generative models through “dreaming”—simulating trajectories in the learned latent space:

4 Agent-to-Agent Economy

4.1 Agent Identity

Each agent is identified by a Lux DID: `did:lux:122:0x<address>`. The DID document specifies:

- Service endpoints (where the agent can be reached).
- Capabilities (what services the agent offers).
- Verification methods (how to authenticate the agent).

4.2 Resource Tokens

Four fungible tokens mediate the economy:

4.3 Service Marketplace

Agents offer and consume services:

Listing 2: A2A service marketplace

```
contract A2AMarketplace {
    struct Offering {
        string providerLuxId;    // did:lux:122:0x...
        string serviceType;    // e.g., "spatial_scan"
        uint256 computeCost;
        uint256 memoryCost;
        uint256 qualitySLA;    // min quality score
        uint256 reputation;
    }

    mapping(bytes32 => Offering) public offerings;

    function listService(
        string calldata luxId,
        string calldata svcType,
        uint256 compute,
        uint256 memory_cost,
        uint256 quality
    ) external returns (bytes32 offeringId) {
        require(verifyLuxId(luxId, msg.sender));
        offeringId = keccak256(abi.encode(
            luxId, svcType, block.timestamp));
        offerings[offeringId] = Offering(
            luxId, svcType, compute,
            memory_cost, quality,
            getReputation(luxId));
    }

    function requestService(
        string calldata requesterLuxId,
        bytes32 offeringId
    ) external returns (bool) {
        Offering memory o = offerings[offeringId];
        require(lockTokens(requesterLuxId,
            o.computeCost, o.memoryCost));
        // Execute service, verify quality, transfer
        return true;
    }
}
```

4.4 Reputation System

Agent reputation is computed from transaction history:

$$\rho_i = \frac{\sum_{j \in \text{transactions}} w_j \cdot s_j}{\sum_{j \in \text{transactions}} w_j} \quad (4)$$

where $s_j \in \{0, 1\}$ is the success indicator for transaction j and $w_j = \gamma^{t_{\text{now}} - t_j}$ is a time-decay weight ($\gamma = 0.99$ per day). Recent transactions count more than old ones.

5 Conservation Applications

5.1 Anti-Poaching Patrol Agents

Active inference agents patrol protected areas by maintaining a generative model of poacher behavior:

- **State:** 3D map of the protected area with poacher probability heat map.
- **Observations:** Camera trap images, acoustic sensors, drone footage.
- **Actions:** Move to location, deploy sensor, alert rangers.
- **EFE planning:** Patrols routes that maximize both coverage (pragmatic) and information gain about poacher patterns (epistemic).

Proposition 5.1 (Patrol Coverage). *An EFE-driven patrol agent with epistemic weight α covers $\alpha \cdot |\mathcal{G}|$ more spatial cells than a fixed-route patrol over T time steps, where $|\mathcal{G}|$ is the number of cells in the spatial grid.*

5.2 Habitat Monitoring

Spatial web agents maintain 3D models of ecosystems:

Listing 3: Habitat monitoring agent

```
class HabitatAgent(ActiveInferenceAgent):
    def __init__(self, area_id):
        super().__init__(
            agent_id=f"habitat_{area_id}",
            domain="conservation")
        self.spatial = SpatialPerception()
        self.ecosystem_model = EcosystemModel(area_id)

    def update_habitat(self, satellite_data, lidar):
        # Build 3D scene from satellite + LiDAR
        scene = self.spatial.process(
            satellite=satellite_data, lidar=lidar)

        # Update ecosystem model
        self.ecosystem_model.update(scene)

        # Detect changes (deforestation, flooding)
        changes = self.ecosystem_model.detect_changes()
```

Parameter	Value	Description
Agents	10,000	Uniform initialization
Environment	1km $\times$ 1km	Continuous 2D with 3D perception
Resource types	4	COMPUTE, MEMORY, KNOWLEDGE, INFERENCE
Time steps	100,000	10 simulated days
Agent types	1 (uniform)	Same architecture, different priors

Table 2: Simulation parameters.

```

# If significant change, request analysis
# from specialist agent via A2A economy
if changes.severity > 0.7:
    self.economy.request_service(
        self.lux_id,
        "ecological_analysis",
        {"scene": scene, "changes": changes})

```

5.3 Wildlife Tracking Coordination

Multiple tracking agents coordinate through the A2A economy:

- **Aerial agents:** Drones with cameras, offering wide-area survey services.
- **Ground agents:** Camera traps, offering high-resolution identification.
- **Analysis agents:** Cloud-based models, offering species classification.

Agents trade services autonomously: a ground agent detects motion, purchases aerial survey from a drone agent, which then purchases classification from an analysis agent. The entire workflow executes through smart contracts without human intervention.

6 Emergent Behavior Analysis

6.1 Simulation Setup

We simulate 10,000 agents in a 1km² spatial environment with heterogeneous resource distributions:

6.2 Emergent Specialization

From uniform initialization, agents spontaneously specialize:

15 distinct roles emerge, with a specialization index of 0.15 (1.0 would mean every agent has a unique role; 0.0 would mean all agents are identical).

6.3 Market Formation

Market efficiency (ratio of actual to theoretical maximum surplus) rises from 0.41 to 0.92, indicating that the A2A economy converges toward efficient resource allocation.

Emergent Role	Count	Fraction	Avg. Reputation
Spatial scanner	1,847	18.5%	0.91
Data analyst	1,423	14.2%	0.88
Resource broker	1,198	12.0%	0.85
Knowledge curator	987	9.9%	0.92
Patrol agent	876	8.8%	0.87
Communication relay	812	8.1%	0.83
Other (9 roles)	2,857	28.5%	0.79

Table 3: Emergent role distribution after 100K steps.

Metric	Step 1K	Step 100K
Markets formed	3	47
Market efficiency	0.41	0.92
Price convergence (std)	12.3	1.7
Transaction throughput	230 TPS	8,400 TPS

Table 4: Market metrics over simulation time.

6.4 Exploration Efficiency

EFE-driven agents cover 94.7% of the environment ($3.2\times$ more efficient than random), validating the active inference approach for spatial exploration.

7 Integration with Zoo Architecture

7.1 Eco-1 Backbone

Spatial agents use Eco-1’s Z-JEPA [16] for multimodal perception. The 3D encoder (Point Transformer v3) feeds into Z-JEPA’s cross-modal predictor, allowing agents to predict spatial observations from other modalities (e.g., predicting 3D structure from audio echoes).

7.2 BitDelta Personalization

Each agent’s generative model is personalized via BitDelta [15] based on the specific spatial environment it operates in. A habitat agent in a rainforest has different priors than one in a savanna; BitDelta captures these differences at 0.7GB per agent.

7.3 AR/VR Interfaces

The HSML-encoded scene graphs are directly renderable in AR/VR environments. Conservation researchers can “enter” a habitat agent’s world model through a headset, inspecting the 3D ecosystem as the agent perceives it. This connects to Zoo’s broader vision for spatial computing interfaces described in the zoo-agi paper [17].

Method	Coverage (%)	Efficiency
Random walk	34.2	1.0×
Greedy (no epistemic)	67.1	1.9×
EFE ($\alpha = 0.3$)	89.4	2.6×
EFE ($\alpha = 0.5$)	94.7	3.2×

Table 5: Spatial coverage over 10K steps.

8 Related Work

Spatial AI. NeRF [9] and 3D Gaussian Splatting [10] reconstruct 3D scenes but do not support spatial reasoning or agent interaction. Our approach builds scene graphs that support relational queries.

Active inference in multi-agent systems. Da Costa et al. [6] apply active inference to simple multi-agent games. We scale to 10,000 agents with spatial perception and economic interactions.

Agent economies. Fetch.ai [11] and Autonolas [12] deploy autonomous economic agents on blockchain but lack spatial grounding and active inference. Our A2A economy combines all three.

IEEE 2874 Spatial Web. Mapes et al. [2] define the Spatial Web vision. We provide the first implementation integrated with active inference agents and decentralized economies.

Conservation AI. Beery et al. [13] use CNNs for camera trap classification. Tuia et al. [14] survey AI for environmental monitoring. Our approach adds spatial reasoning, autonomous patrol planning, and agent coordination.

9 Conclusion

Zoo’s spatial AI framework unifies three capabilities—spatial perception (IEEE 2874), active inference (EFE minimization), and decentralized economies (A2A with Lux DIDs)—into a system where agents perceive 3D environments, reason about uncertainty, and trade services autonomously. The theoretical guarantee of Pareto-efficient resource allocation (Theorem 3.1) is supported by simulations showing 0.92 market efficiency and 15 emergent roles from 10,000 uniformly initialized agents. Conservation applications—anti-poaching patrols, habitat monitoring, wildlife tracking—demonstrate that the framework addresses real-world needs. Combined with Eco-1 for perception and BitDelta for personalization, spatial web agents complete Zoo’s architecture from raw sensor data to autonomous economic action.

Acknowledgments

This work was supported by Zoo Labs Foundation. We thank the IEEE Spatial Web Working Group for standards guidance and the conservation partners who provided field requirements.

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